FRONTEND DEVELOPMENT

What is React?

A powerful **JavaScript library** for building modern user interfaces — especially single-page applications (SPAs).

Component-Based

Build encapsulated components that manage their own state, then compose them to make complex UIs.

Efficient UI Updates

React focuses on efficiently updating and rendering the right components when your data changes — keeping interfaces fast and responsive.

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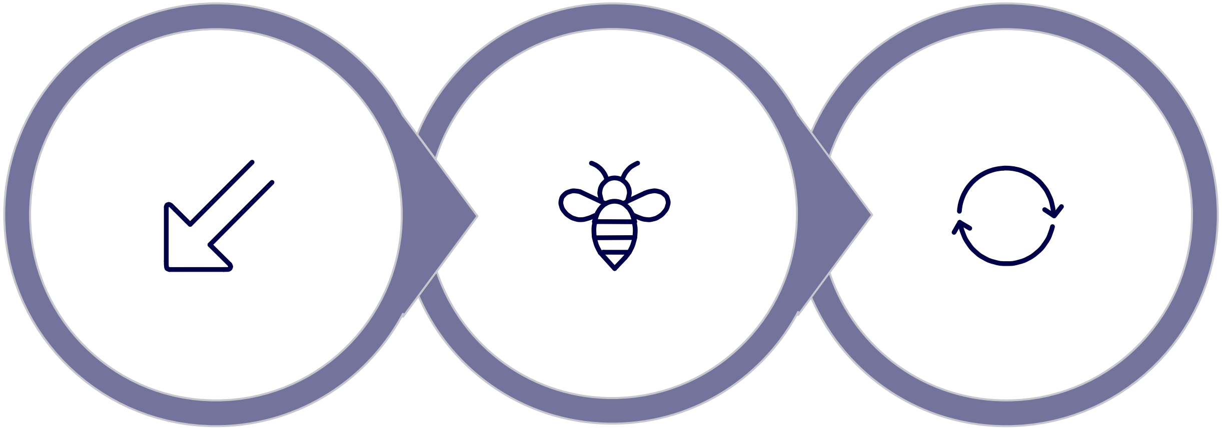
CORE CONCEPT

Single Page Application (SPA)

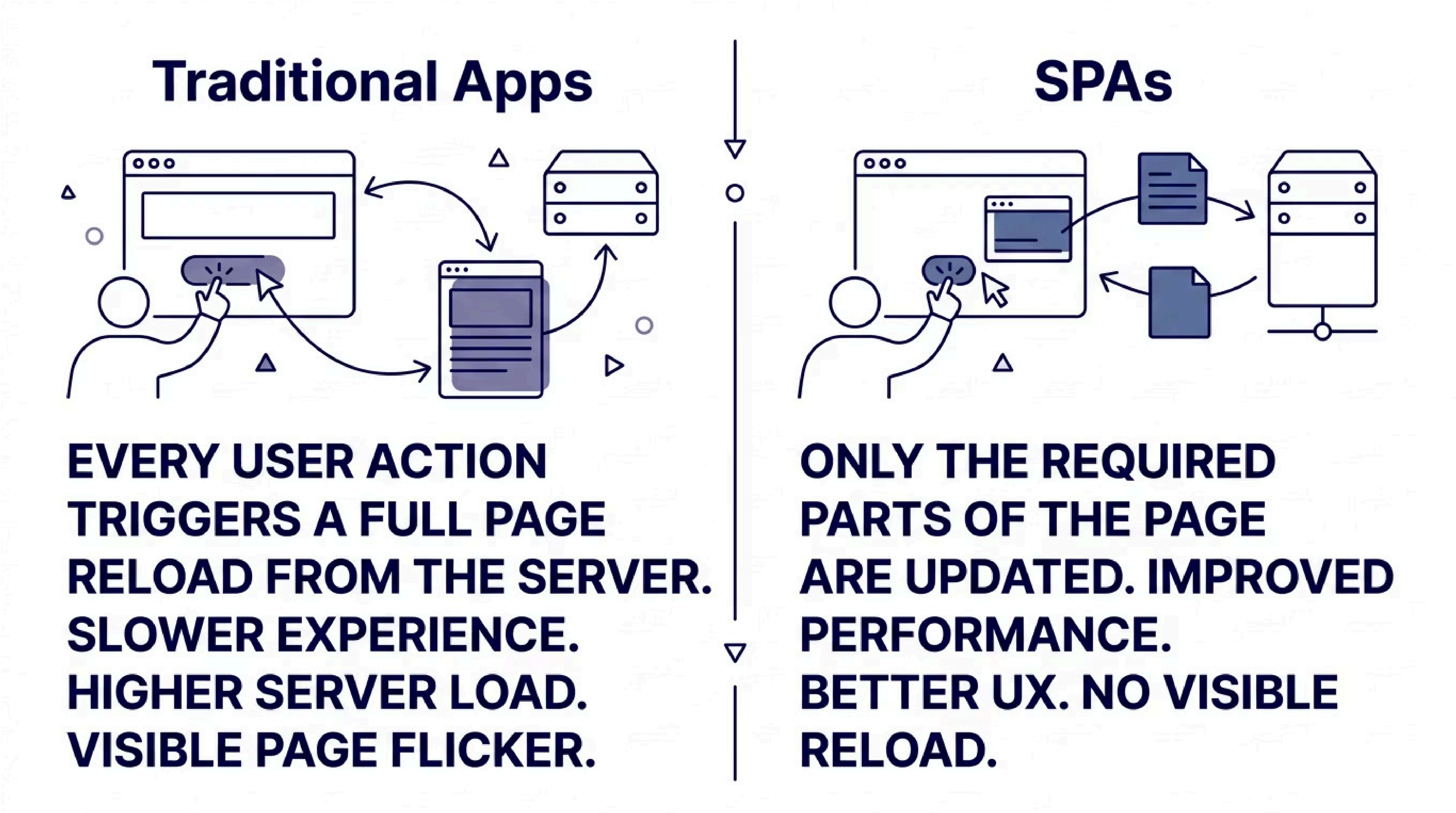
The entire app runs on a **single HTML page** — no full reloads, just seamless, dynamic updates.



User Action **Fetch Data** **Update View**

SPAs communicate with the backend via REST APIs to fetch and send data dynamically, delivering a faster and smoother user experience compared to traditional web applications.

SPA vs Traditional Applications



The shift from traditional multi-page applications to SPAs represents a fundamental change in how web apps deliver user experiences — moving rendering logic to the client and reducing unnecessary server round-trips.

REAL-WORLD EXAMPLES

Examples of SPAs in the Wild

Some of the most widely used applications on the internet are built as SPAs — demonstrating the power and scalability of this architecture.

Google Maps

Updates the map view, directions, and location data **dynamically without a full page reload** — delivering a seamless navigation experience.

Gmail & Yahoo Mail

UI updates — reading emails, composing, archiving — all happen **inline without navigating away**, keeping the experience fluid and instant.

REACT'S PURPOSE

The Role of React

Partial Page Updates

Enables efficient partial page updates — the core behaviour that powers SPA architecture. Only the components that need to change are re-rendered.

State-Driven Rendering

Handles UI rendering based on **state changes** — when data updates, React reactively re-renders the affected parts of the interface automatically.

Full-Stack Integration

Integrates seamlessly with backend services such as **Spring Boot** — enabling complete full-stack application development with clean API communication.

INDUSTRY ADOPTION

Who Uses React?

React was created by **Facebook** and has since become one of the most widely adopted frontend libraries in the world — trusted by major companies for production-grade applications.

Facebook

The original creator of React — built and battle-tested at massive scale across billions of users.

WhatsApp

Uses React for its web interface — delivering real-time messaging updates efficiently.

Dropbox

Relies on React for its web application — managing complex UI state across a large user base.

React is not just a trend — it is a proven, production-grade technology adopted by some of the largest engineering teams in the world.

Why Not Vanilla JavaScript?

Vanilla JavaScript is entirely possible for small projects — but as applications grow in complexity, the limitations become significant.

The Problem at Scale

- Manual DOM manipulation becomes repetitive and error-prone
- Data binding must be handled entirely by hand
- Difficult to maintain consistency across a large codebase
- No built-in component structure — code becomes tangled

How React Solves This

- Abstracts away complex DOM update logic
- Provides a declarative, component-based model
- Improves developer productivity significantly
- Frameworks like React are the industry standard for a reason

> React and similar frameworks don't replace JavaScript — they **abstract its complexity**, letting developers focus on building features rather than managing the DOM.

KEY VALUE

Why React Matters

React delivers a compelling combination of developer productivity, scalability, and user experience — making it the go-to choice for modern web development.

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Library by Facebook

Backed by one of the world's largest engineering organisations

∞

Scalable Architecture

Designed for large teams and complex, long-lived applications

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Full Page Reloads

Enables modern SPAs with zero unnecessary page refreshes

→ **Simplifies UI Development**

Handles the heavy lifting of DOM updates — developers declare *what* the UI should look like, React figures out *how* to get there.

→ **Easier Maintenance**

Component-based architecture makes large codebases easier to navigate, test, and maintain — even across big teams.

→ **Modern User Experiences**

Enables responsive, reactive interfaces that meet the expectations of today's users — fast, fluid, and engaging.